

Day 7: Git Version Control & Containerization

Git Workflows, Traditional Deployment Bottlenecks, and Docker Basics

Module 1: Git Workflows & Traditional Hurdles

Git is a distributed version control system that tracks source code modifications during development. It allows teams to work on the same code simultaneously, manage history branches, and rollback changes.

Traditional Deployment Bottlenecks

- **Configuration Drift:** 'It works on my machine' problems. Differences in library versions and OS configurations between local systems and production servers.
- **Resource Waste:** Virtual machines require a full Guest Operating System, consuming gigabytes of storage and memory resources.
- **Slow Startup Time:** Booting virtual machines takes minutes because the guest OS must initialize system services.

Git Bash to GitHub CLI Push Sequence

```
git init
git add .
git commit -m 'Initial commit'
git branch -M main
git remote add origin <github-repository-url>
git push -u origin main
```

Module 2: Containerization & Docker Foundations

Containerization is a form of OS-level virtualization. Containers share the host OS kernel, making them lightweight, fast to start, and portable across platforms.

Feature	Virtual Machines (VM)	Containers (Docker)
OS Architecture	Requires a full Guest OS on a hypervisor	Shares host OS kernel; no guest OS needed
Size Footprint	Large (Gigabytes)	Very Small (Megabytes)
Startup Time	Minutes (full OS boot process)	Seconds (starts like a system process)
Overhead	High (CPU/RAM dedicated to guest OS)	Minimal (resources allocated to app only)

Dockerfile Reference Directives

- **FROM:** Specifies the base parent image (e.g. FROM ubuntu:22.04).
- **RUN:** Executes shell commands during the image build phase to install packages.
- **COPY:** Transfers local code folders from the host computer into the container filesystem.
- **EXPOSE:** Informs Docker that the container listens on specified network ports at runtime.
- **CMD:** Defines the default command executed when the container starts running.